

Carlos Rodriguez Dumar

Frontend Developer | React | Next.js | Typescript | Agile | AI Interface Developer | 15+ Years in Software

 carlosdumar4@gmail.com |  [Portfolio](#) |  [Github](#) |  Medellín, Colombia

Summary

Senior Frontend Engineer with 15+ years in software development and 5+ years specializing in React, Next.js, and TypeScript. Recently built a production AI chat interface serving thousands of daily users within a native iOS application at EPAM, implementing Pub/Sub communication patterns and Content Security Policy hardening. Experienced delivering accessible, high-performance web applications across global consultancies (EPAM, Globant, Endava) for US enterprise clients in pharma, education, and healthcare.

Key Highlights

- **AI Chat Interface** - Built a real-time AI-powered chat UI embedded in a native iOS app, serving thousands of daily pharmaceutical users at EPAM.
 - **Performance Engineering** - Improved Lighthouse scores from ~60 to ~80 and reduced CLS, TTI, and initial bundle size on a large-scale education platform at Globant.
 - **Accessibility Champion** - Delivered WCAG AA - compliant interfaces tested with NVDA screen readers and ARIA attributes for education products used by teachers and students.
 - **Critical Bug Reduction** - Led code quality initiatives that eliminated ~60% of critical bugs through code reviews, standards enforcement, and testing practices.
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Technical Skills

Core: Javascript (ES6+), Typescript, React 18, [Next.js](#) (SSR, SSG, ISR, App Router), HTML5, CSS3

State Management: Redux, Redux Toolkit, MobX, Zustand, Tanstack Query, Context API

Styling: TailwindCSS, Styled Components, CSS Modules, SASS

Testing: Jest, React Testing Library

Build & Tooling: Webpack, Git, Jenkins, CI / CD, ESLint, Prettier

Cloud & Deployment: AWS (S3), GCP, Vercel, Netlify

Accessibility: WCAG 2.1 AA, NVDA, ARIA, Semantic HTML

Other: Storybook, Pub/Sub Pattern, Content Security Policy (CSP), XSS Prevention, REST APIs

Languages: Spanish (Native), English (Professional working proficiency - all projects with US clients conducted in English), Portuguese (Intermediate)

Professional Experience

EPAM - Medellín, Colombia

Senior Software Engineer Jan. 2025 – August 2025

Built a dynamic AI chat interface embedded in a native iOS application for the pharmaceutical industry. The AI assistant enabled healthcare professionals (HCPs) and pharmaceutical representatives to schedule calls and manage interaction lists through conversational UI, serving thousands of daily users.

- Developed the frontend chat application using React and Typescript, embedded within the iOS app via WebView using iframe integration.
 - Implemented the Publisher/Subscriber (Pub/Sub) communication pattern to enable secure, bidirectional messaging between the web-based chat UI and the native iOS application layer.
 - Enforced Content Security Policy (CSP) headers and XSS prevention strategies to secure the embedded web application within the iOS environment.
 - Integrated Redux and Redux Toolkit for scalable state management of conversation threads, user sessions, and HCP scheduling workflows.
 - Collaborated with a team of 3 frontend engineers alongside backend and mobile teams to establish reliable communication with the AI model API.
 - Wrote unit and integration tests using Jest and React Testing Library to ensure reliability across the chat interface.
 - Gained hands-on experience with iOS Simulator for debugging and testing the embedded WebView experience.
 - **Technologies:** React, Typescript, Redux/Toolkit, Jest, React Testing Library, Jenkins, iOS WebView.
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GLOBANT - Medellín, Colombia

Web UI Developer June 2021 – Jan. 2025

Worked across two majors projects for US-based education clients over 3.5 years, delivering accessible, performant web applications using React, Next.js, and Typescript.

Project: Digital Teacher Planning & Guidance Platform Built a large-scale web application that digitized teacher guidance materials for the education industry, using Next.js with both SSR and SSG rendering strategies across two interconnected applications with a shared custom component library.

- Developed responsive, WCAG AA-compliant interfaces using React, Next.js, Typescript, and MobX for reactive state management.
- Utilized both Server-Side Rendering (SSR) and Static Site Generation (SSG) in [Next.js](#) to optimize page load performance and SEO across the platform.
- Consumed and extended a Storybook-based custom component library (~10 shared components), and built 4-5 additional application-specific components for the teacher planning views.
- Delivered full WCAG AA accessibility compliance, testing with NVDA screen reader and implementing semantic HTML and ARIA attributes – critical for an education product used by teachers and students.
- Executed performance profiling that improved Lighthouse scores from ~60 to ~80, reducing Cumulative Layout Shift (CLS), Time to Interactive (TTI), and initial bundle weight through code splitting and optimization workflows triggered on commit.
- Led code reviews and authored coding standards documentation adapted by the project team, contributing to a ~60% reduction in critical bugs across the application.
- **Technologies:** React, [Next.js](#) (SSR/SSG), Typescript, MobX, Storybook, Styled Components, Jest, WCAG AA, NVDA.

Project: ePub Screen Reader Application Developed a web-based reader application that parsed and rendered .epub file content for accessible on-screen reading.

- Built the epub parsing and content rendering interface using React and Javascript.
 - Ensured accessible reading experience aligned with the broader platform's WCAG standards
 - Technologies: React, Javascript, HTML5, CSS3.
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ENDAVA – Medellín, Colombia

Senior Technician Dec. 2019 – May. 2021

- Developed fully functional responsive prototypes using React, Redux, and Typescript based on business requirements for US clients.
- Built a threaded comment section for a healthcare company's web portal, implementing nested comment threads, user interactions, and content moderation UI.
- Tested frontend code across multiple browsers to ensure cross-browser compatibility
- Worked in both Agile and Waterfall environments across all stages of the product lifecycle
- **Technologies:** React, Typescript, Redux, Webpack, Jest

Technician April. 2017 – Dec. 2019

- Transitioned from jQuery / .Net development to React during this period, adopting modern frontend architecture and component-based design patterns.
 - Designed, coded, tested, and documented UI components as part of larger engineering teams
 - Produced interactive prototypes and contributed to the design of system components
 - **Technologies:** Javascript, jQuery, React, .NET (C#), CSS3, HTML5, SASS, Grunt.
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INTERGRUPO – Medellín, Colombia

Development Engineer Aug. 2011 – Aug. 2016

- Implement test-driven development practices to validate software implementations before internal release.
 - Analyze functional artifacts and requirements to deepen understanding of problems and solutions before coding.
 - Constructed and modified software according to specifications, design guidelines, and defined tooling standards.
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Education

Santo Tomas University, Medellin, Colombia Feb. 2004 – Aug. 2009 BS, Telecommunication Engineer

Personal Projects

- **Portfolio Site** - React-based personal portfolio deployed on Netlify with CI/CD
(<https://carlosrodriguez-portfolio.netlify.app/>)
- **MammeyTube** - Video platform project showcasing React development skills
(<https://github.com/adminDreamBike/mameytube>)